

Web Engineering & Development (SWE 363)

4.1 JavaScript Fundamentals

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Announcements

- Project Assignment #1 is due This Thursday at 11:59 PM
- Quiz 1 next class
- Individual assignments released
- Demo submission, commit each TODO

In today's lecture:

- Objects
- Maps
- String
- Date
- Math
- Exception Handling
- Regular Expressions

Reference:

- Zybook: 4.9 to 4.15

JavaScript Objects

Key Concepts:

- **Object Creation**: Literal notation vs constructor
- **Properties**: Accessing and modifying object properties
- **Methods**: Functions as object properties
- **this keyword**: Context and binding
- **Object iteration**: `for...in` loops
- **Object methods**: `Object.keys()`, `Object.values()`, `Object.entries()`

Objects - Creation & Access

```
// Object literal notation
const student = {
  name: "Ahmed",
  age: 20,
  major: "Computer Science",
  greet: function() {
    return `Hello, I'm ${this.name}`;
  }
};

// Accessing properties
console.log(student.name);           // "Ahmed"
console.log(student["age"]);         // 20
console.log(student.greet());        // "Hello, I'm Ahmed"

// Adding new properties
student.gpa = 3.8;
```

Objects - Methods & **this**

```
const calculator = {  
  result: 0,  
  add: function(num) {  
    this.result += num;  
    return this;  
  },  
  multiply: function(num) {  
    this.result *= num;  
    return this;  
  },  
  getResult: function() {  
    return this.result;  
  }  
};  
// Method chaining  
calculator.add(5).multiply(3).add(2);  
console.log(calculator.getResult()); // 17
```

Maps

Key Concepts:

- **Map vs Object:** When to use each
- **Map methods:** `set()`, `get()`, `has()`, `delete()`, `clear()`
- **Map iteration:** `for...of` loops
- **Map size:** `size` property
- **Key types:** Any type can be a key

Maps - Creation & Usage

```
// Creating a Map
const studentGrades = {
  "Ahmed": 95,
  "Sara", 87,
  "Omar", 92
};

// Accessing values
console.log(studentGrades.get("Ahmed")); // 95
console.log(studentGrades.has("Sara")); // true

// Map size
console.log(studentGrades.size); // 3
```


String Objects

Key Concepts:

- **String methods:** `length`, `charAt()`, `indexOf()`, `substring()`
- **String manipulation:** `toUpperCase()`, `toLowerCase()`, `trim()`
- **String searching:** `includes()`, `startsWith()`, `endsWith()`
- **String splitting:** `split()`, `join()`
- **Template literals:** Backticks and `${}` syntax

String Objects - Basic Methods

```
const text = "  Hello, World!  ";

// Length and character access
console.log(text.length);           // 17
console.log(text.charAt(0));        // " "
console.log(text[0]);               // " " (bracket notation)

// Case conversion
console.log(text.toUpperCase());     // "  HELLO, WORLD!  "
console.log(text.toLowerCase());     // "  hello, world!  "

// Trimming whitespace
console.log(text.trim());            // "Hello, World!"
```

String Objects - Advanced Methods

```
const sentence = "JavaScript is awesome and powerful";

// Finding positions
console.log(sentence.indexOf("is"));           // 11
console.log(sentence.lastIndexOf("a"));        // 20

// Extracting substrings
console.log(sentence.substring(0, 10));        // "JavaScript"
console.log(sentence.slice(-7));               // "powerful"

// Splitting and joining
const words = sentence.split(" ");
console.log(words); // ["JavaScript", "is", "awesome", "and", "powerful"]

const joined = words.join("-");
console.log(joined); // "JavaScript-is-awesome-and-powerful"
```

Date Objects

Key Concepts:

- **Date creation:** `new Date()`, `new Date(year, month, day)`
- **Date methods:** `getFullYear()`, `getMonth()`, `getDate()`, `getDay()`
- **Time methods:** `getHours()`, `getMinutes()`, `getSeconds()`,
`getMilliseconds()`
- **Date formatting:** `toString()`, `toISOString()`, `toLocaleDateString()`

Date Objects - Creation & Basic Methods

```
// Creating dates
const now = new Date();
const specificDate = new Date(2024, 8, 17); // September 17, 2024
const dateFromString = new Date("2024-09-17");

console.log(now); // Current date and time
console.log(specificDate); // September 17, 2024

// Getting date components
console.log(now.getFullYear()); // 2024
console.log(now.getMonth()); // 8 (September, 0-indexed)
console.log(now.getDate()); // 17
console.log(now.getDay()); // 2 (Tuesday, 0=Sunday)
```

Date Objects - Formatting

```
const date = new Date(2024, 8, 17, 14, 30, 0);

// Date formatting
console.log(date.toDateString());           // "Tue Sep 17 2024"
console.log(date.toISOString());           // "2024-09-17T14:30:00.000Z"
console.log(date.toLocaleDateString());     // "9/17/2024" (locale dependent)
```

Math Objects

Key Concepts:

- **Mathematical constants:** `Math.PI`, `Math.E`
- **Rounding methods:** `Math.round()`, `Math.floor()`, `Math.ceil()`, `Math.trunc()`
- **Power and roots:** `Math.pow()`, `Math.sqrt()`
- **Trigonometric functions:** `Math.sin()`, `Math.cos()`, `Math.tan()`
- **Random numbers:** `Math.random()`
- **Min/Max:** `Math.min()`, `Math.max()`

Math Objects - Constants & Rounding

```
// Mathematical constants
console.log(Math.PI);           // 3.141592653589793
console.log(Math.E);           // 2.718281828459045

// Rounding methods
const num = 4.7;
console.log(Math.round(num));   // 5
console.log(Math.floor(num));   // 4
console.log(Math.ceil(num));    // 5
console.log(Math.trunc(num));    // 4

// Rounding to decimal places
const price = 19.99;
const rounded = Math.round(price * 100) / 100;
console.log(rounded);           // 19.99
```


Math Objects - Power, Roots & Random

```
// Power and roots
console.log(Math.pow(2, 3));           // 8
console.log(Math.sqrt(16));            // 4

// Random numbers
console.log(Math.random());             // Random number between 0 and 1

// Random integer between min and max
function randomInt(min, max) {
    return Math.floor(Math.random() * (max - min + 1)) + min;
}
console.log(randomInt(1, 10));          // Random integer between 1 and 10

// Min and Max
console.log(Math.min(5, 2, 8, 1));      // 1
console.log(Math.max(5, 2, 8, 1));      // 8
```

Exception Handling

Key Concepts:

- **try-catch-finally blocks**: Error handling structure
- **Throwing errors**: `throw` statement

Exception Handling - Basic Structure

```
try {
    const result = 10 / 0;
    console.log(result);
} catch (error) {
    console.log("An error occurred:", error.message);
}

function divide(a, b) {
    try {
        if (b === 0) { throw new Error("Division by zero is not allowed"); }
        return a / b;
    } catch (error) {
        console.log("Error:", error.message);
        return null;
    } finally {
        console.log("Division operation completed");
    }
}
```

Regular Expressions

Key Concepts:

- **Regex syntax:** Patterns, flags, and metacharacters
- **Character classes:** `[a-z]` , `[0-9]` , `[^abc]`
- **Quantifiers:** `*` , `+` , `?` , `{n}` , `{n,m}`
- **Anchors:** `^` , `$` , `\b`
- **Special characters:** `\d` , `\w` , `\s` , `.`
- **Regex methods:** `exec()` , `match()` , `replace()`

Regular Expressions - Basic Patterns

```
// Creating regex
const pattern1 = /hello/;
const pattern2 = new RegExp("hello");

// Basic matching
const text = "Hello, world!";
console.log(pattern1.test(text));           // true
console.log(/world/.test(text));           // true

// Case insensitive flag
const caseInsensitive = /hello/i;
console.log(caseInsensitive.test("HELLO")); // true

// Global flag
const globalPattern = /o/g;
console.log(text.match(globalPattern));    // ["o", "o"]
```

Regular Expressions - Char. Classes & Quantifiers

```
const text = "Contact us at info@example.com or call 123-456-7890";

// Character classes
const emailPattern = /[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}/;
console.log(emailPattern.test(text));    // true

// Quantifiers
const phonePattern = /\d{3}-\d{3}-\d{4}/;
console.log(phonePattern.test(text));    // true

// Word boundaries
const wordPattern = /\b\w{4}\b/g;
console.log(text.match(wordPattern));    // ["info", "call"]

// Special characters
const digitPattern = /\d+/g;
console.log(text.match(digitPattern));    // ["123", "456", "7890"]
```

Regular Expressions - Practical Examples

```
// Email validation
function isValidEmail(email) {
    const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    return emailRegex.test(email);
}

console.log(isValidEmail("user@example.com")); // true
console.log(isValidEmail("invalid-email"));    // false

// Phone number formatting
function formatPhone(phone) {
    const phoneRegex = /(\d{3})(\d{3})(\d{4})/;
    return phone.replace(phoneRegex, "($1) $2-$3");
}

console.log(formatPhone("1234567890")); // "(123) 456-7890"
```

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Demo

Starter code at: [web-engineering-kfupm-classroom-4-2-js-advance-4-2-js-advance-Dromarjh-main](#)

Next Class

- Quiz